

OMKAR THOMBRE

Adelaide, SA 5000, Australia | +61 433 237 995 | omkararun.thombre@adelaide.edu.au

[LinkedIn](#) | [GitHub: Omkarth](#) | [Portfolio](#)

CAREER OBJECTIVE

Master of Computer Science student at Adelaide University (graduating July 2027) with a primary focus on cybersecurity and applied AI. Built CyberLLM, a 350M-parameter security-domain language model trained from scratch, and a maritime dark-ship detection pipeline using SAR imagery and deep learning. Currently a Teaching Assistant for cybersecurity coursework. Seeking summer internships in cybersecurity, SOC analysis, or AI-driven security engineering where I can ship production-grade tools.

EDUCATION

Master of Computer Science

Jul 2025 – Present

Adelaide University, Adelaide, SA, Australia | GPA: 6.667 / 7.0

Bachelor of Engineering in Computer Engineering

Nov 2021 – May 2025

MGM College of Engineering & Technology (University of Mumbai), Navi Mumbai, Maharashtra, India | CGPA: 7.75 / 10

EXPERIENCE

Teaching Assistant — Security Operations & Incident Response (INFO 2042)

Feb 2026 – Present

Adelaide University | Adelaide, SA, Australia

- Deliver weekly practical sessions for tutorial group PR03 (~18 students), one of several cohorts within a 112-student cybersecurity course, running hands-on SOC triage simulations, incident response walkthroughs, and recovery/crisis communication exercises.
- Produced supplementary teaching materials including print-ready student worksheets, colour-coded tutor answer keys, and a Weeks 5–10 progress tracking spreadsheet aligned with student RangeForce platform activity.
- Facilitate group-based tabletop incident-response exercises across Weeks 9–10, guiding student teams through scenario response and session reporting as part of a 3-stage assessment culminating in individual reflective submissions.
- Provide one-on-one academic support and run doubt-clearing sessions covering threat detection, log analysis, and incident handling procedures, working under course coordinator Dr. Xiaogang Zhu.

Data Science Intern — Academor (in partnership with Flutura)

Jan 2023 – Feb 2023

Remote, India

- Completed a 2-month industry-partnered data science internship program co-branded with Flutura Decision Sciences, covering supervised learning, time-series forecasting, and exploratory data analysis using Python, scikit-learn, and pandas.
- Built an end-to-end classification project on a self-selected dataset, performing EDA, feature engineering, model training, and accuracy evaluation as the capstone deliverable.
- Strengthened applied ML foundations across linear/logistic regression, SVM, and FB Prophet forecasting through structured weekly modules and live mentor-led sessions.

PROJECTS

CyberLLM — Domain-Specific Cybersecurity Language Model

2025 – 2026

- Pretrained a **350M-parameter** (303M exact) LLaMA-3 style decoder-only transformer entirely from scratch on **5B tokens** of cybersecurity-weighted data (~90% security content from Primus-FineWeb, MITRE ATT&CK, NVD/CVE, NIST SP 800, and security Stack Exchange), implementing grouped-query attention (GQA), SwiGLU, RMSNorm, and rotary positional embeddings (RoPE).
- Designed a custom 32K SentencePiece BPE tokenizer with domain-specific vocabulary covering CVE identifiers, MITRE ATT&CK technique IDs, and security tooling, improving tokenization efficiency over general-purpose tokenizers for security text.
- Fine-tuned the pretrained base on 3,750 curated cybersecurity instruction–response pairs over 3 epochs, achieving a **validation loss of 1.28** (perplexity 3.7) and producing coherent responses across incident response, SQL injection, ransomware handling, and zero-trust architecture queries.
- Engineered a Flask REST API with SOC-facing endpoints (alert analysis, playbook generation, contextual chat) and a React + Vite SOC Assistant dashboard featuring severity-filtered alert triage, per-alert AI threat analysis, MITRE ATT&CK mapping, raw log inspection, and an “Ask CyberLLM” Q&A panel; the full stack runs locally on Apple M4 MPS with zero cloud inference dependency.
- Completed end-to-end training across two model iterations (125M and 350M) for **~\$61 USD (~\$91 AUD)** total compute on RunPod (NVIDIA A100 + A40), demonstrating that domain-specialised LLMs are achievable on student budgets.
- Tech:** Python, PyTorch, SentencePiece, Flask, React, Vite, Hugging Face Tokenizers, RunPod (A100, A40 GPUs), Weights & Biases.

Dark Ship Detection — Maritime Surveillance via SAR + AIS Fusion

2025 – 2026

- Built an end-to-end deep learning pipeline for detecting vessels that disable AIS transponders to evade tracking (“dark ships”), fusing Sentinel-1B dual-polarisation SAR imagery (25,306 × 16,644 pixel scene over Iceland/Greenland) with AIS broadcast data to identify illegal fishing tied to the global **\$23B IUU industry**.
- Processed raw SAR GRD products into 3,978 training tiles via rasterio windowed reading (RAM-safe at 12GB), composing VV, VH, and VV/VH ratio channels into pseudo-RGB and applying log-dB transforms with percentile stretching; diagnosed and patched

a silent label-corruption bug in the SARFish dataset (swapped top/bottom bbox coordinates) producing 0.000 mAP despite normal-looking training curves.

- Trained YOLOv8s (11.1M params, anchor-free, Distribution Focal Loss) with SAR-specific augmentation and integrated SAHI sliced inference (320×320 sub-tiles, NMS merge at IoU 0.4) — increasing detections **309× from 2 to 619** raw candidates and demonstrating that a tuned 11M-parameter model with smart inference outperformed a custom 75M-parameter YOLOv8s-P2 variant that failed to converge after 2 hours of training.
- Achieved **0.270 mAP@50** (precision 0.41, recall 0.30) across four iterative versions (0.000 → 0.053 → 0.142 → 0.270); geo-referenced detections via 210 GCP-based pixel-to-lat/lon interpolation and matched against AIS using Haversine distance (5 km / ±30 min window), classifying **124 of 161 vessel candidates (77%) as dark ships**.
- Delivered the full pipeline on free-tier infrastructure (Google Colab T4, 15GB Drive), with novel contributions including SAHI applied to dual-polarisation SAR and a 5 km Haversine matching threshold as a practical alternative to Doppler phase-refocusing approaches.
- **Tech:** Python, Ultralytics YOLOv8, SAHI, rasterio, OpenCV, NumPy, Google Colab (T4), Google Drive.

SOCP — Secure Chat Protocol & Adversarial Peer Review (Team Project, 4 members)

2025 – 2026

- Contributed to a WebSocket-based secure messaging system implementing the cohort-designed SOCP v1.3 protocol with RSA-4096 end-to-end encryption and canonical JSON signature verification, as part of a 4-person Masters security project; the implementation interoperated successfully with 3 other groups' independent codebases.
- Architected the server backbone and asynchronous routing layer in Python asyncio — 3-introducer bootstrap network, hash-based duplicate-suppression cache with TTL cleanup, and 15s/45s heartbeat liveness — and administered the team GitHub repository.
- Spearheaded the intentional-backdoor strategy, planting three deliberately-hidden vulnerabilities (an undocumented DEBUG_COMMAND bypassing signature checks, hardcoded admin credentials in USER_HELLO, and selective signature-verification skipping); **only 3 of ~13 peer-auditor groups detected the stealthiest backdoor**, validating the visibility-vs-stealth tradeoff.
- Conducted adversarial security audits of three peer groups' implementations, identifying SQL injection in user lookup, MD5 password hashing without KDF, authentication timing oracles, introducer-pinning bypass, and TOCTOU race conditions — using static analysis, protocol-aware fuzzing (Hypothesis), timing-attack t-tests, and MITM simulations, with remediation diffs and proof-of-concept payloads.
- **Tech:** Python, WebSockets, asyncio, RSA-4096 (OAEP + PSS), canonical JSON, SQLite, Git/GitHub.

NumCompute — Vectorised Scientific Computing Toolkit (Team Project, 4 members)

2026

- Contributed to a NumPy-only ML utility library implementing 40+ functions across 10 modules (data I/O, preprocessing, sort/search, ranking, statistics, evaluation metrics, numerical optimisation, pipeline orchestration) as part of a 4-person Masters coursework team for Programming for AI.
- Owned three modules end-to-end: chunked-streaming CSV loaders with NaN handling (io.py), feature scalers/imputers/one-hot encoder built on NaN-aware vectorised reductions (preprocessing.py), and numerically stable shared utilities including overflow-safe sigmoid/softmax, logsumexp, and distance helpers (utils.py).
- Authored the project's unit test suite covering NaN propagation, tie handling under multiple ranking conventions, division-by-zero in scaling utilities, and shape-validation edge cases, contributing to **89% line coverage** (516 statements, 37 uncovered) measured via pytest-cov.
- Coordinated collaboration through a branch-per-developer GitHub workflow with PR-based reviews; team aligned on consistent fit/transform/predict API contracts before implementation to reduce cross-module integration friction.
- **Tech:** Python, NumPy, pytest, pytest-cov, GitHub, flake8.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, SQL, HTML/CSS, Bash

ML / AI: PyTorch, NumPy, Hugging Face Tokenizers, SentencePiece, Ultralytics (YOLOv8), SAHI, scikit-learn, OpenCV, rasterio, MATLAB

Frameworks: React, Vite, Next.js, Node.js, Express, Flask, Tailwind CSS, Bootstrap, shaden/ui, WebSockets, asyncio

Tools & Platforms: Git/GitHub, VS Code, Postman, Docker, Google Colab, RunPod, Vercel, Figma, Linux, pytest, pytest-cov, flake8, Weights & Biases, RangeForce

Security & Domains: SOC Operations, Incident Response, Threat Detection, MITRE ATT&CK, RSA-4096 (OAEP + PSS), Canonical JSON, Static Analysis, Protocol Fuzzing (Hypothesis), MITM Simulation, Computer Vision, NLP, SAR / Remote Sensing, Full-Stack Development

CERTIFICATIONS

- Google Cloud Computing Foundations — Google Cloud Skills Boost
- RangeForce Community Edition — Cybersecurity Training
- ISO 9001 Lead Implementer — Quality Certification